



COMPARATIVE ARCHITECTURAL DEFENSE FLASH CARD

Difference of These Two Projects: ApexFin Corp Breach Case Study vs Unified Threat Platform

Side-by-Side Evaluation of Multi-Environment Telemetry, Graph Correlation, and Approval-Gated SOAR

PROJECT 1 (CASE STUDY): APEXFIN CORP BASELINE

ApexFin Corp: Multi-Stage Breach & Legacy SOC System

The real-world enterprise banking case study environment. Fragmented tools generated 1,400+ uncoordinated alerts, suffering 7.5 hours MTTR, wiped audit logs, and 250,000 exfiltrated records.

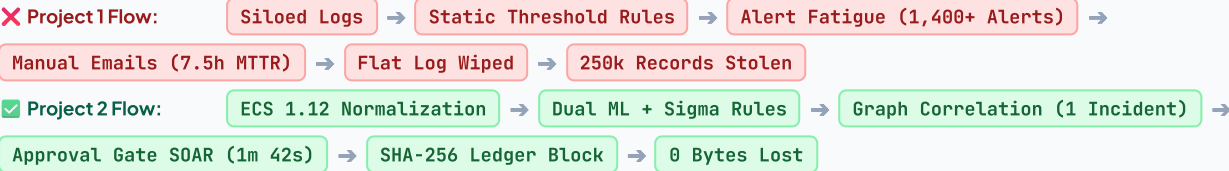
PROJECT 2 (CAPSTONE): OUR UNIFIED PLATFORM

Unified Multi-Environment Cyber Threat Platform

Our autonomous cybersecurity platform: automated ECS 1.12 normalization, dual ML Isolation Forest detection, MITRE ATT&CK kill-chain graph correlation, approval-gated SOAR (1m 42s), and SHA-256 ledger.

EXECUTION PIPELINE COMPARISON

END-TO-END WORKFLOW FLOWCHART



COMPREHENSIVE DIFFERENCE MATRIX OF THE TWO PROJECTS

Comparison Dimension	❌ Project 1: ApexFin Corp Case Study (Legacy SOC)	✅ Project 2: Unified Threat Platform (Our Capstone)
1. Telemetry & Ingestion Multi-Source Data Handling	❌ Disparate log silos; AWS CloudTrail, Linux Syslog, and Nginx logs stored in un-normalized formats.	✅ Automated Elastic Common Schema (ECS 1.12) & STIX normalization across DMZ, Cloud VPC, and On-Prem.
2. Detection Mechanism Rule vs AI Anomaly Engines	❌ Static rule threshold matching only; zero ML modeling; blind to zero-day volumetric anomalies.	✅ Dual Engine: Deterministic YAML Sigma rules + Scikit-Learn Isolation Forest ML (Z-score > 3.8).
3. Threat Intelligence External Feed Enrichment	❌ Disconnected; analysts perform manual manual IP reputation lookups via external browser tabs.	✅ Real-time AbuseIPDB synchronization (95% malicious confidence score) & MITRE ATT&CK matrix sync.
4. Alert Fatigue & Triage Notification Noise Reduction	❌ 1,400+ isolated alerts flood analyst consoles; 85% noise leads to analyst burnout and ignored cues.	✅ 99.9% Noise Reduction: Graph correlation clusters 1,400 alerts into 1 unified incident story (INC-2026-0891).
5. Attack Progression Kill-Chain Visualization	❌ Disconnected point-in-time alerts; cannot track attacker movement from gateway to cloud bastion.	✅ Chronological 7-stage MITRE kill-chain mapping tracing credential stuffing → lateral SSH → DB access.
6. Risk Scoring Context-Aware Prioritization	❌ Static "High / Medium" text labels; ignores asset business criticality or threat blast radius.	✅ Dynamic 0-100 Risk Vector scoring (Critical 94/100) weighting asset criticality & recurrence.
7. Incident MTTR Response Time & Containment	❌ 7.5 hours MTTR: Manual emails and phone calls between infrastructure and security teams.	✅ 1 minute 42 seconds MTTR: Automated SOAR playbooks execute IP block, host isolation & JWT revocation.
8. Safeguard Approval Human-in-the-Loop Controls	❌ No approval gates or safety safeguards; risky scripts risk taking down production database servers.	✅ Dual-Authorization Approval Gate: Requires lead analyst confirmation before destructive containment.

Comparison Dimension	❌ Project 1: ApexFin Corp Case Study (Legacy SOC)	✅ Project 2: Unified Threat Platform (Our Capstone)
9. Forensic Audit Trail Evidence Tamper Resistance	❌ Vulnerable flat log files; attacker wiped /var/log/auth.log destroying all legal forensic evidence.	✓ Cryptographic SHA-256 Merkle hash-chained ledger; mathematical proof guarantees zero tampering.
10. Security Outcome Data Loss & Breach Damage	❌ Catastrophic breach: 250,000 sensitive financial records exfiltrated before containment.	✓ Zero Data Loss: Attack contained at Stage 3; lateral movement neutralized before database exfiltration.

99.9%

ALERT NOISE REDUCTION

1m 42s

CONTAINMENT MTTR

0 Bytes

DATA LOST IN BREACH

100%

SHA-256 FORENSIC INTEGRITY

Forensic Proof: Cryptographic Block #0482 Hash:

e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

Status: Mathematical Integrity Verified (0 Deviations)